

Attention Beyond the Training Length

How Score Scaling Controls Length Generalization in a Reduced Binary Attention Classifier

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Motivation

Training on short sequences does not guarantee reliable performance on much longer inputs [2]. As sequence length grows, the target must compete with more non-target tokens for the same attention budget.

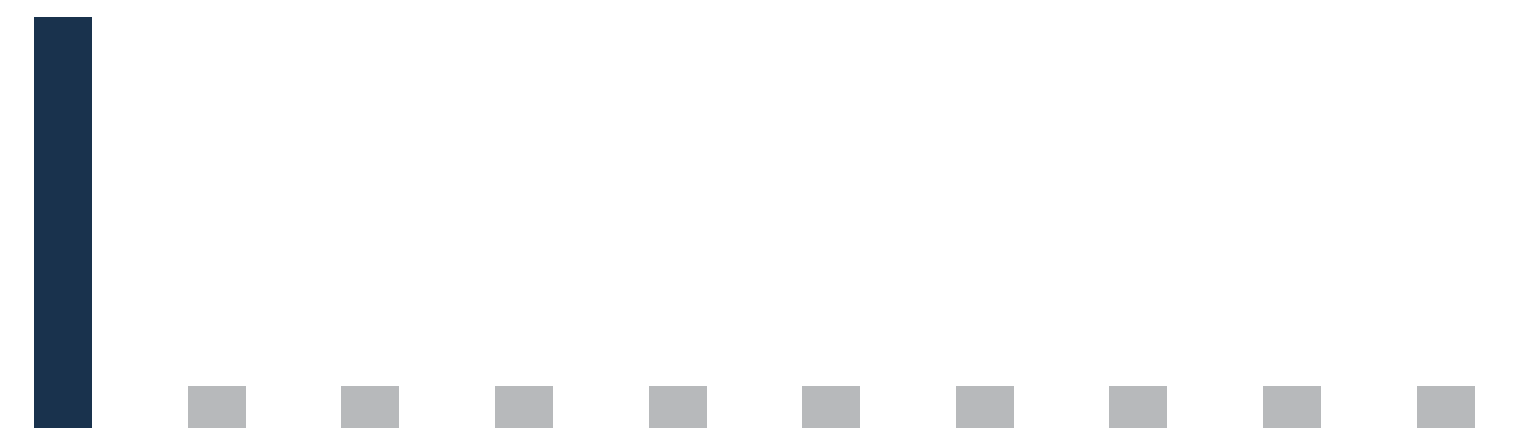
How must attention sharpen with length to remain concentrated on the target?

Why Fixed Attention Dilutes with Length

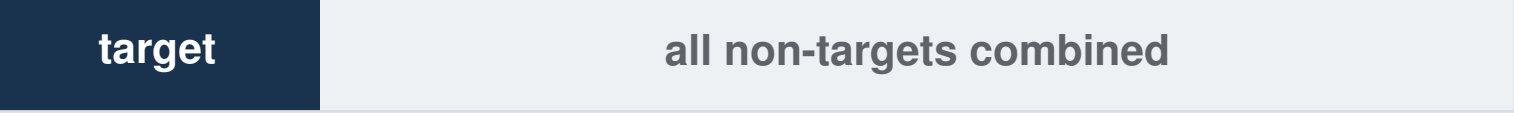
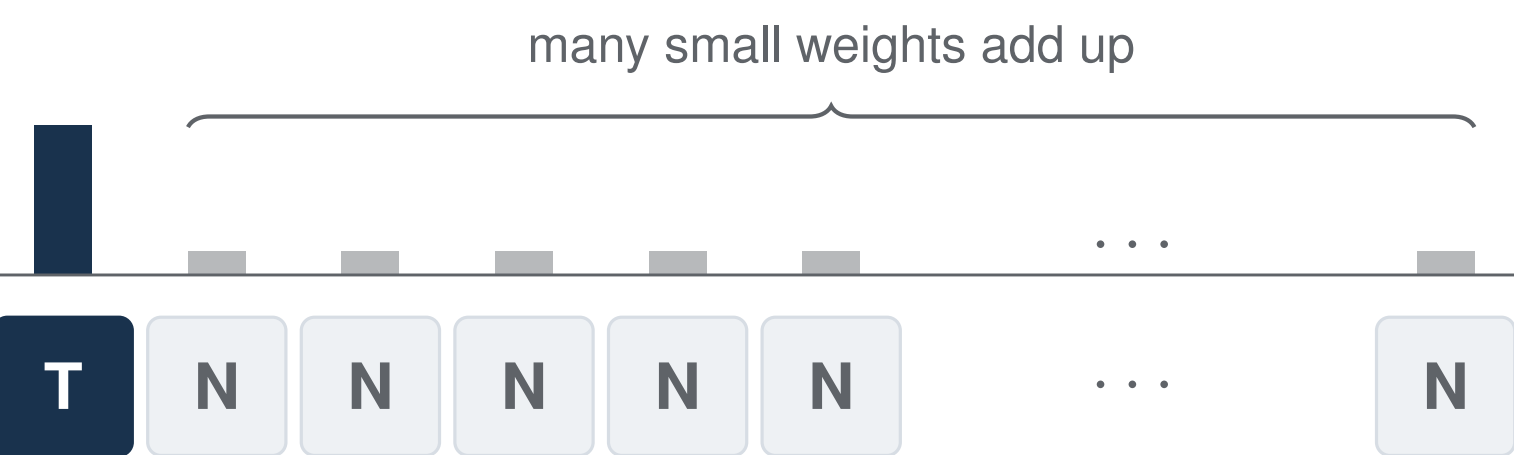
The model detects whether a designated target token is present in the sequence.

Short sequence ($n = 10$)

attention per token



Long sequence



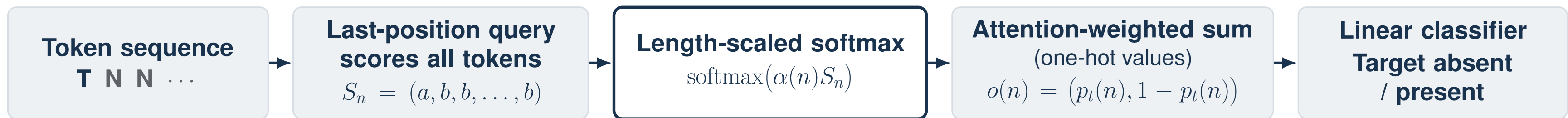
(schematic)

A fixed score margin separates the target from each non-target, but cannot offset their growing aggregate weight.

Let $p_t(n)$ be the target's share of attention. With target score a and $n - 1$ identical non-target scores b , standard softmax attention [1] gives

$$p_t(n) = \frac{e^{\Delta}}{e^{\Delta} + (n-1)} \rightarrow 0 \text{ as } n \rightarrow \infty, \quad \Delta = a - b > 0.$$

How Length-Aware Scaling Counters Dilution



Before softmax, every score is multiplied by a positive factor $\alpha(n)$: the ranking is unchanged, but the effective score margin grows from Δ to $\alpha(n)\Delta$.

Constant

$$\alpha(n) = 1$$

No response to sequence length
 $p_t(n) \rightarrow 0$

Log

$$\alpha(n) = \log n$$

Preset logarithmic growth
 $p_t(n) \rightarrow 1$ if $\Delta > 1$

Learned log

$$\alpha(n) = 1 + c \log(1 + n)$$

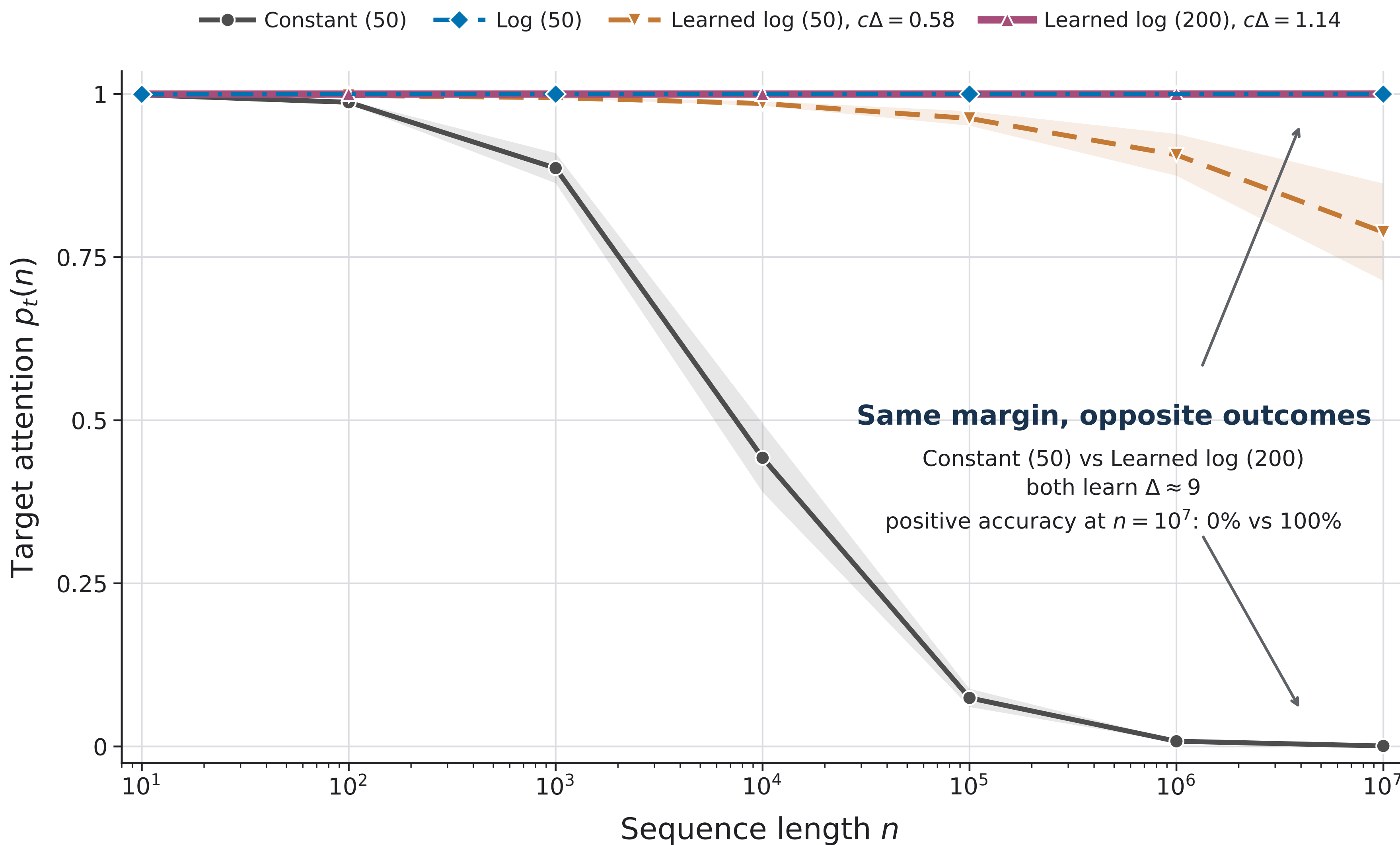
Optimization learns $c > 0$
 $p_t(n) \rightarrow 1$ if $c\Delta > 1$

Asymptotic criterion $p_t(n) \rightarrow 1 \iff \alpha(n)\Delta - \log(n-1) \rightarrow +\infty$.

The effective score margin must outgrow the logarithm of the number of non-target competitors.

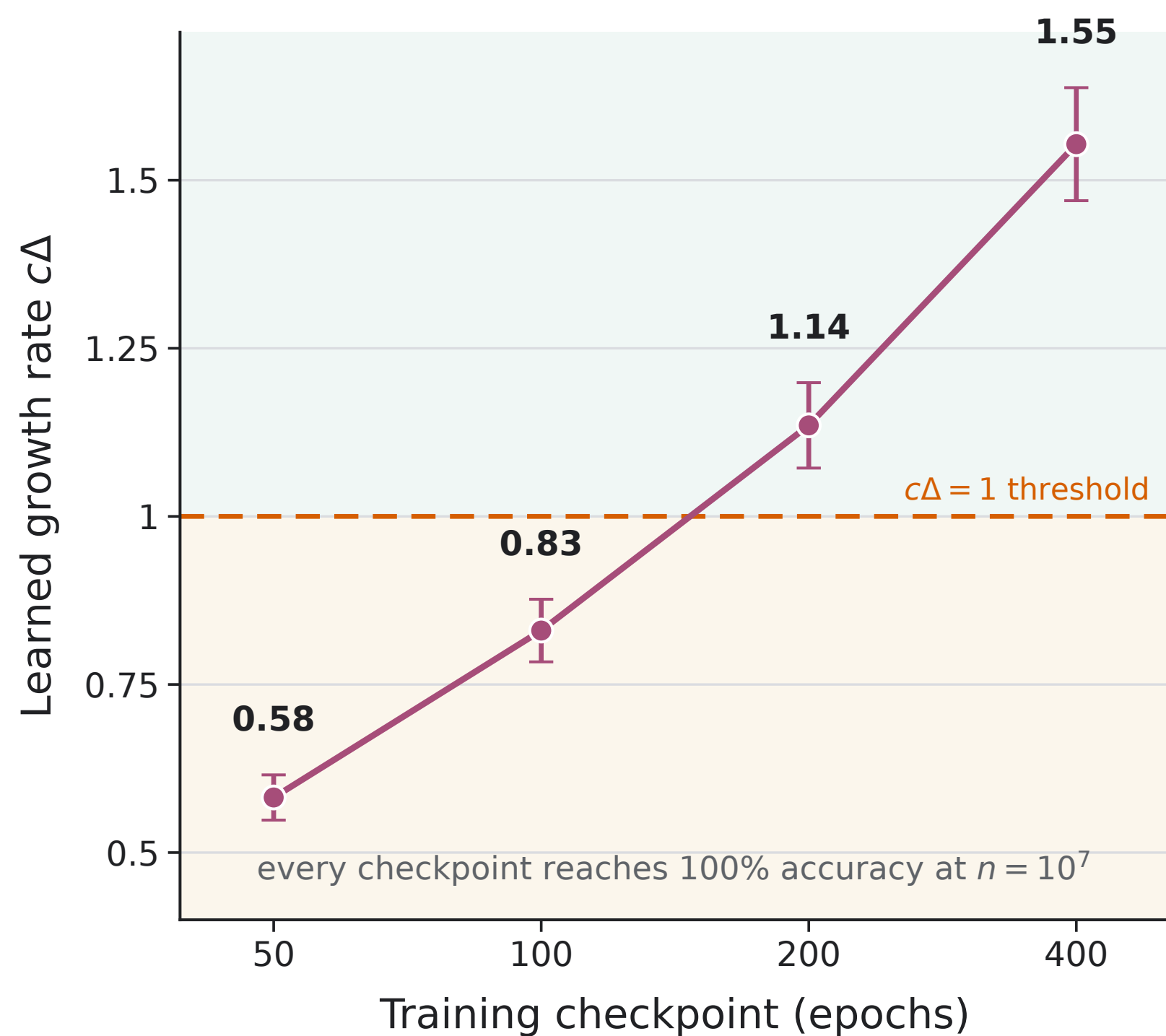
Same Training-Length Accuracy, Different Long-Length Behavior

TRAIN: $n = 10$ | EVALUATE: $n = 10$ to 10^7 | 5 RANDOM SEEDS



Means over five seeds; bands show ± 1 s.d.

Learned-Log Threshold



Means over five seeds; error bars show ± 1 s.d.

A Finite Pass Can Hide Later Failure. Checkpoints below $c\Delta = 1$ pass the $n = 10^7$ benchmark yet are predicted to fail at greater lengths.

Key Takeaways & Scope

1 Fixed scaling postpones, but cannot prevent, target-attention dilution.

2 Sufficiently strong logarithmic sharpening can prevent dilution.

Target attention converges to one when $\Delta > 1$ (Log) or $c\Delta > 1$ (Learned log).

Scope. These thresholds are exact for the reduced two-score classifier studied here; whether analogous conditions hold in full transformers remains open.

References & Acknowledgments

[1] Vaswani et al. (2017). *Attention Is All You Need*. NeurIPS.
[2] Press et al. (2022). *Train Short, Test Long: Attention with Linear Biases Enables Input Length Extrapolation*. ICLR.

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